

```
import pandas as pd
import numpy as np

ds = pd.read_csv('wan_dataset.csv')
```

ACTIVITY 1

```
print("Frequency: ", ds["timestamp"].value_counts())
print("Mean: ", ds["timestamp"].mean())
print("Median: ", ds["timestamp"].median())
print("Mode: ", ds["timestamp"].mode())
print("Standard Deviation: ", ds["timestamp"].std())
print("Variance: ", ds["timestamp"].var())
print("Interquartile range: ", ds["timestamp"].quantile(0.75) - ds["timestamp"].quantile(0.25))
```

```
Frequency: timestamp
0      10
1      10
2      10
3      10
4      10
..
195    10
196    10
197    10
198    10
199    10
Name: count, Length: 200, dtype: int64
Mean: 99.5
Median: 99.5
Mode: 0      0
1      1
2      2
3      3
4      4
...
195    195
196    196
197    197
198    198
199    199
Name: timestamp, Length: 200, dtype: int64
Standard Deviation: 57.748744217769485
Variance: 3334.9174587293646
Interquartile range: 99.5
```

ACTIVITY 2

```
sum = 0

for x in ds["packet_loss"]:
    sum = sum + x

mean = sum/len(ds["packet_loss"])

print("Mean: ", mean)
```

```
Mean: 0.9810000000000001
```

```
print("Mode: ")

frequency = ds["packet_loss"].value_counts()

for x in frequency:
    if(x == max(frequency)):
        print(frequency[frequency==x].index)
        break
```

```
Mode:
Index([ 1.2, 0.4, 1.6, 1.76, 0.8, 0.28, 1.48, 0.68, 1.88, 1.08, 0.56, 0.84,
        0.96, 0.16, 1.36, 1.24, 0.04, 0.44, 1.64, 1.96, 1.12, 0.32, 1.52, 0.72,
        1.92, 1.4, 0.6, 1.8, 1.0, 0.2, 1.68, 0.88, 0.08, 1.28, 0.48, 1.84,
        1.16, 0.36, 1.56, 0.76, 0.24, 1.44, 0.64, 0.92, 1.04, 0.52, 1.72, 1.32,
        0.12],
      dtype='float64', name='packet_loss')
```

```

n = len(ds["packet_loss"])

data = sorted(ds["packet_loss"])

if n % 2 == 0:
    median = (data[n//2 - 1] + data[n//2]) / 2
else:
    median = data[n//2]

print("Median: ", median)

```

Median: 1.0

```

sum = 0

for x in ds["packet_loss"]:
    sum = sum + x

mean = sum / len(ds["packet_loss"])

sum_square = 0

for x in ds["packet_loss"]:
    sum_square = sum_square + (x - mean) ** 2

variance = sum_square / (len(ds["packet_loss"]) - 1)

standard_deviation = variance ** 0.5

print("Variance: ", variance)
print("Standard Deviation: ", standard_deviation)

```

Variance: 0.33340570285142657
Standard Deviation: 0.5774129396293666

```

data = sorted(ds["packet_loss"])

n = len(data)

if n % 2 == 0:
    lower = data[:n//2]
    upper = data[n//2:]
else:
    lower = data[:n//2]
    upper = data[n//2 + 1:]

m = len(lower)

if m % 2 == 0:
    q1 = (lower[m//2 - 1] + lower[m//2]) / 2
else:
    q1 = lower[m//2]

m = len(upper)

if m % 2 == 0:
    q3 = (upper[m//2 - 1] + upper[m//2]) / 2
else:
    q3 = upper[m//2]

iqr = q3 - q1

print("Q1: ", q1)
print("Q3: ", q3)
print("Interquartile Range: ", iqr)

```

Q1: 0.48
Q3: 1.48
Interquartile Range: 1.0

ACTIVITY 3

```
import math

gd = pd.read_csv("packet_loss_grouped.csv")

print(gd)
```

	packet_loss_range	frequency
0	0-0.2	199
1	0.2-0.4	200
2	0.4-0.6	200
3	0.6-0.8	200
4	0.8-1.0	200
5	1.0-1.2	200
6	1.2-1.4	200
7	1.4-1.6	200
8	1.6-1.8	200
9	1.8-2.0	201

```
midpoints = []

for value in gd["packet_loss_range"]:
    limits = value.split("-")

    lower = float(limits[0])
    upper = float(limits[1])

    midpoint = (lower + upper) / 2

    midpoints.append(midpoint)

gd["midpoint"] = midpoints
gd["cumulative_frequency"] = gd["frequency"].cumsum()

print(gd)
```

	packet_loss_range	frequency	midpoint	cumulative_frequency
0	0-0.2	199	0.1	199
1	0.2-0.4	200	0.3	399
2	0.4-0.6	200	0.5	599
3	0.6-0.8	200	0.7	799
4	0.8-1.0	200	0.9	999
5	1.0-1.2	200	1.1	1199
6	1.2-1.4	200	1.3	1399
7	1.4-1.6	200	1.5	1599
8	1.6-1.8	200	1.7	1799
9	1.8-2.0	201	1.9	2000

```
frequency = gd["frequency"]

print("Frequency:")
print(frequency)
print("Total Frequency =", frequency.sum())
```

```
Frequency:
0    199
1    200
2    200
3    200
4    200
5    200
6    200
7    200
8    200
9    201
Name: frequency, dtype: int64
Total Frequency = 2000
```

```
mean = (gd["midpoint"] * gd["frequency"]).sum() / frequency.sum()

print("Mean: ", mean)
```

```
Mean:  1.0009000000000001
```

```
freq = gd["frequency"].max()

for x in range(len(gd)):
    if gd["frequency"][x] == freq:
```

```
print("Highest Frequency: ", freq)
print("Modal class: ", gd["packet_loss_range"][x])
```

```
Highest Frequency: 201
Modal class: 1.8-2.0
```

```
N = gd['frequency'].sum()

median_position = N/2

for x in range(len(gd)):
    if gd["cumulative_frequency"][x] >= median_position:

        median_class = gd['packet_loss_range'][x]

        limits=median_class.split("-")
        l=float(limits[0])
        h=float(limits[1]) - l

        if x!=0:
            cf = gd['cumulative_frequency'][x-1]

        f = gd['frequency'][x]

median = l + h*((median_position - cf)/f)

print("Median: ", median)
```

```
Median: 1.0049751243781095
```

```
for x in gd['midpoint']:
    s = (x-mean)**2

variance = s/len(gd['frequency'])

std = math.sqrt(variance)

print("Standard Deviation: ", std)

print("Variance: ", variance)
```

```
Standard Deviation: 0.2843203844257389
Variance: 0.08083808099999996
```

```
N = gd['frequency'].sum()

Q1_position = N / 4
Q3_position = 3 * N / 4

for x in range(len(gd)):
    if gd["cumulative_frequency"][x] >= Q1_position:

        Q1_class = gd['packet_loss_range'][x]

        limits = Q1_class.split("-")
        l1 = float(limits[0])
        h1 = float(limits[1]) - l1

        if x == 0:
            cf1 = 0
        else:
            cf1 = gd['cumulative_frequency'][x - 1]

        f1 = gd['frequency'][x]

        break

Q1 = l1 + h1 * ((Q1_position - cf1) / f1)

for x in range(len(gd)):
    if gd["cumulative_frequency"][x] >= Q3_position:

        Q3_class = gd['packet_loss_range'][x]
```

```
limits = Q3_class.split(" ")
l3 = float(limits[0])
h3 = float(limits[1]) - l3

if x == 0:
    cf3 = 0
else:
    cf3 = gd['cumulative_frequency'][x - 1]

f3 = gd['frequency'][x]

break

Q3 = l3 + h3 * ((Q3_position - cf3) / f3)

IQR = Q3 - Q1

print("Q1:", Q1)
print("Q3:", Q3)
print("Interquartile Range:", IQR)
```

```
Q1: 0.501
Q3: 1.501
Interquartile Range: 0.9999999999999999
```